

FR-017 — Agent Evaluation & Observability

FR017 — `sources/acceptance.md`

Masterclass source package snapshot

Functional Requirement	FR-017
Edition	Engineering Learning Academy — Source Package snapshot
Status	**Complete / Frozen / Accepted
Audience	Academy generation and study (snapshot; SoT remains authoritative)
Document type	Source package snapshot (PDF)

Faithful rendering of a regenerable Masterclass source snapshot. Do not hand-edit; regenerate via `python scripts/build_masterclass_package.py`. Repository documentation remains the source of truth.

Contents

FR-017 — Agent Evaluation & Observability

Status: Complete / Frozen / Accepted

Date: 2026-08-07

Documentation close-out: 2026-08-07 (M4 freeze); repository consistency close-out confirmed same day

Recommendation: ACCEPT AND FREEZE — FR-017 COMPLETE

Binding product posture: narrow derive-only evaluation / learning substrate

Engineering Learning Academy: Ready — canonical engineering record = this report; attachable package = [masterclass/FR017/](#) (README.md , MANIFEST.md , regenerable sources/)

Next: Horizon 1B (FR-018+) may proceed when the owner chooses — **not gated on FR-017**. Ordinary prep remains `cic agent run`. Do not auto-start 1B in this close-out.

ADR: [ADR-009](#) (Accepted — M1–M4 close-out)

Milestones (historical records): [M0](#) (Accepted), [M1](#) (Accepted), [M2](#) (Accepted — GO), [M3](#) (Accepted), [M4](#) (this freeze).

This document is the **canonical engineering record** for FR-017. Milestone reports remain historical; do not reopen exit criteria without owner request.

1. Executive Summary

FR-017 delivers a **narrow derive-only orchestration evaluation capability**:

```
Existing OrchestrationRun / Handoff / child AgentRun
↓
Deterministic metric derivation
↓
R1-R12 reconstructability evaluation
↓
Read-only owner CLI
↓
Operational / learning insight
```

Layer	Delivered
Contracts	<code>multi_agent.observability</code> (M1)
Corpus	15/15 deterministic cases — GO (M2)
CLI	<code>cic agent orchestrate metrics</code> / <code>metrics-corpus</code> (M3)
Freeze	M4 acceptance + Academy package

Near-term commercial value is low. Daily preparation is unchanged (`cic agent run`). FR-017 does **not** justify dashboards or live telemetry at this stage. Value is **audit reconstructability**, **honest missing-data semantics**, and **interview- transferable evaluation engineering**.

Horizon 1B remains unblocked.

2. Business / engineering problem

After FR-016, orchestration audits exist but owners lacked a first-class way to:

1. reconstruct the authority story without reading source;
2. compare runs deterministically offline;
3. distinguish missing provider metadata from measured zero;
4. detect orphan / contradictory parent-child linkage.

FR-017 answers those needs **without** inventing a second SoT or an observability product.

3. Final architecture

Concern	Owner	FR-017 role
Workflow traces	FR-008	None
Truth engine	FR-014	Cite status only
BOPA metrics	FR-015	Reuse <code>AgentRunMetrics</code>
Orchestration audits	FR-016	Source records
Derive + R1-R12 + CLI	FR-017	Evaluation only

Invariant: No DOS / BOPA / OBS / Handoff / AgentRun runtime changes; no new events; no telemetry store; no dashboard.

Package: `career_intelligence.multi_agent.observability` (+ corpus + presentation)

CLI: `cic agent orchestrate {metrics, metrics-corpus}` (+ existing FR-016 commands)

4. Final corpus results

Metric	Value
Cases	15/15 passed
<code>go_no_go</code>	GO
Deterministic repeat	True
Runtime instrumentation required	False

Case	Result	Notes
C01 complete successful	PASS	Full R1–R12
C02 delegation blocked	PASS	deny + stop
C03 BOPA child	PASS	provider/tokens present
C04 OBS brief	PASS	brief goal
C05 prepare_then_brief	PASS	both specialists
C06 missing optional metadata	PASS	None preserved
C07 measured zero	PASS	<code>0</code> $\neq$ missing
C08 orphaned child	PASS	R11 FAIL expected
C09 missing child join	PASS	ids correlate; tokens missing
C10 stale/incomplete handoff	PASS	reconstructable stale path
C11 safe resume	PASS	R12 + single visit
C12 loop stops	PASS	repeated / circular / no_progress
C13 provider unavailable	PASS	fail-closed stop
C14 contradictory audit	PASS	R7+R11 FAIL expected
C15 mixed aggregate	PASS	corpus roll-up

Harness: `run_observability_corpus()` / `cic agent orchestrate metrics-corpus`.

5. R1–R12 results (normative behaviour)

Id	Question (summary)	PASS means	FAIL means
R1	Owner goal	Goal kind + opportunity present	Missing goal identity
R2	Observed state	<code>last_observation</code> + hash	No observation evidence
R3	Selection	Allow/deny handoff or selection events	No selection evidence yet
R4	DelegationPolicy	Allow/deny fields consistent	Inconsistent policy fields
R5	Authority boundary	Target $\in$ {obs, bopa}	Unknown target
R6	Handoff lifecycle	Acceptance present	Missing acceptance
R7	Child output	Completed allow cites expected child (stopped may omit)	Missing child refs
R8	Stop reason	Present when not running	Terminal without stop
R9	Owner next	Present when awaiting_owner	Awaiting without action
R10	Limits	Limits visible	Invalid limit config
R11	Parent/child walk	Correlation complete (or vacuous empty)	Orphans / gaps
R12	Resume/ idempotency	Hashes and/or keys (or early vacuous)	No resume evidence

Rules: PASS requires evidence; FAIL names the gap; contradictions are surfaced (C08/C14), never repaired; evaluation is deterministic.

6. Owner manual validation

Script: `scripts/run_fr017_m4_manual.py` — **PASSED** (2026-08-07).

Demo	Evidence
A	Persisted <code>prepare_then_brief</code> under disposable store; metrics CLI reconstructs full story; JSON unchanged
B	<code>--fixture C01</code>
C	<code>--fixture C05</code>
D	<code>--fixture C06 (missing)</code>
E	<code>--fixture C07 (0)</code>
F	<code>--fixture C08 (ORPHAN / R11 FAIL)</code>
G	<code>--fixture C14 (R7+R11 FAIL)</code>
H	<code>metrics-corpus 15/15 GO</code>
I	Read-only proof (A byte-identical; fixtures in-memory)

Persisted reconstruction confirmed markers: goal → observed state → selection → delegation → lifecycle → child → stop → owner next → limits → idempotency → R1–R12.

7. Metadata semantics

Representation	Meaning
<code>None / display missing</code>	Absent optional provider/model/token/cost/latency
<code>0 / 0.0</code>	Measured zero
Count <code>0</code>	Empty measurable count (handoffs, events, ...)

Never coerce missing → zero. C06/C07/C09/C15 prove the distinction.

8. Safety / read-only verification

- Metrics CLI loads only (`load` / `load_handoff` / optional agent `load`)
- Fixtures and corpus perform no store writes
- No DOS start/resume, no BOPA/OBS execution from FR-017 paths
- No pipeline mutation, submission, discovery, or truth waiver
- Unit proof: `test_cli_metrics_from_store_read_only`

9. Product-value assessment (honest)

Question	Answer
Improve daily preparation?	No material improvement — prefer <code>cic agent run</code>
Materially improve Horizon 1B?	No — 1B not dependent on FR-017
Useful operational insight?	Yes — reconstruct / debug orchestration audits
Improve audit/debugging?	Yes — R1–R12 + orphans + missing≠zero
Justify dashboards/telemetry now?	No
Remain narrow evaluation capability?	Yes
Future richer observability?	Only with failing evidence that derive-only cannot answer owner-critical questions at scale

Posture: Low near-term commercial value. High learning / engineering-substrate value.

10. Learning outcomes (Academy)

1. Derive before instrument
 2. Observability ≠ dashboards
 3. Audit reconstructability as acceptance
 4. Missing versus zero
 5. Parent/child correlation and orphan detection
 6. Deterministic offline evaluation
 7. Source-of-truth boundaries (reuse FR-015; do not fork)
 8. Read-only evaluation surfaces
 9. Detect contradictions rather than repair them
 10. Avoid observability theatre
-

11. Technical debt classification

Item	Class
Derive-only metrics + R1–R12 + CLI	Accepted (frozen)
Early-running R3 vacuity	Accepted limitation
Token/cost completeness offline	Deferred (null OK)
Metrics CLI polish / UX	Deferred
Dashboards / alerts / live tracing	Out of Scope
Telemetry backend / monitoring SaaS	Out of Scope
Richer time-travel replay	Future FR (if ever justified)
Fault-injection framework productisation	Deferred (patterns exist in corpus)
Horizon 1B integration	Out of Scope for FR-017; 1B separate

12. Tests

Suite	Path
M1 contracts	tests/unit/multi_agent/test_observability_m1.py
M2 corpus	tests/unit/multi_agent/test_observability_corpus_m2.py
M3 CLI	tests/unit/multi_agent/test_observability_cli_m3.py
FR-016 regression	tests/unit/multi_agent/test_cli_m3.py
FR-015 regression	tests/unit/agent/test_cli.py
Manual M4	scripts/run_fr017_m4_manual.py

13. Documentation / Academy

Frozen navigation across README, AGENTS, roadmap, functional spec, domain model, engineering principles, testing strategy, implementation notes, phase history, changelog **1.110**, ADR-009 close-out.

Academy package: [masterclass/FR017/](#) — regenerate:

```
python scripts/build_masterclass_package.py FR017
```

14. Definition of Done

Criterion	Status
M0–M3 accepted	Yes
ADR-009 accepted	Yes
Final corpus green 15/15	Yes
Owner manual validation	Yes
R1–R12 confirmed	Yes
Missing-versus-zero confirmed	Yes
No runtime mutation / no new SoT	Yes
Product assessment honest	Yes
Learning outcomes captured	Yes
Docs frozen	Yes
FR017 Masterclass package ready	Yes
Horizon 1B unblocked	Yes
No scope leakage	Yes

15. Acceptance recommendation

ACCEPT AND FREEZE — FR-017 COMPLETE

16. Final repository status

Item	Status
FR-001–FR-016	Frozen (unchanged)
FR-017	Complete / Frozen / Accepted
Horizon 1A Stage 11	Closed
Horizon 1B	Not started; not blocked
Daily prep	cic agent run